various points of preadmission, port openings, equalization of cut-offs, release and the closures can be as favorably arranged with Walschaert motion as with the link motion examples presented.

The following tables of weights are given for 22-in. consolidation engines built by the Baldwin Locomotive Works, and said to be identical in everything except the valve motion:

	Complete moving and structural parts		Moving parts only.	
	Stephenson.	Walschaert.	Stephenson.	Walschaert.
Crossheads	676	746	676	746
Guide bearer	814	1,116	...	...
Guides	1,712	1,712	...	...
Eccentrics	600	...	600	...
Crank arms	...	250	...	250
Eccentric straps	1,100	...	1,100	...
Main crank pins	520	516	520	516
Links	238	418	238	418
Reverse shaft	925	655	925	655
Rockers and boxes	618	730	618	730
Rocker rods and hangers	169	...	169	...
Link bearing	...	234	...	...
Eccentric rods	184	264	184	264
Valve rods	220	546	220	546
Valve yokes	154	140	154	140
Valve rod guide	24	28	...	...
Complete set	7,354	8,321	4,804	4,265

While the engine with Walschaert motion weighed 1,000 lbs. more than the other one on account of the motion, yet the weight of the moving parts was less. The *American Engineer and Railroad Journal*, June, 1905, published some figures showing saving of weight by the use of Walschaert motion on L. S. & M. S. engines, as follows: 1,283 lbs. on a consolidation, 1,215 lbs. and 1,745 lbs. respectively on two classes of Prairie type engines. These figures indicate a larger saving than the foregoing example, but it is possible that as these were early developments, they may not be representative of present successful practice. Similar valve motion parts on engines abroad are very much lighter than we dare use in our own practice. It seems fair to conclude, therefore, that we may yet look for improvement in this respect.

The matter of lead has received much attention, particularly with link motion, but, as a matter of fact, the measured full gear lead is only used possibly for a few turns of the wheels in starting the engine, and when running notches are in use the lead is entirely different, the amount being dependent upon various conditions. As the running notch leads for best results are within narrow limits, it is apparent that the full gear leads vary within a wide range. The amount of full gear lead is therefore of little importance in the operation of the engine if the running notch lead is right. If these premises are correct, then there can be no argument as against the constant lead characteristics of the Walschaert motion, provided the lead is the proper amount for the running notches. The valve motion diagrams show that all the other events as derived by a link motion can be duplicated by the Walschaert motion so that, except in so far as the lead is concerned, equivalent operation can be obtained; hence, equivalent economies.

The practical operation of Walschaert motion is best shown by the testimony of roads using it in considerable numbers, and at a recent meeting of motive power officers and locomotive builders held to discuss the results of the use of Walschaert motion engines, the roads represented having about 1,000 such engines, it was the unanimous opinion that Walschaert motion was equally well adapted to fast and slow passenger and freight service; that equivalent economies in fuel and water were obtained; that no reductions of tonnage ratings were necessary; that expense of maintenance and repairs were reduced; that inspection and repairs were facilitated; that construction advantages in the way of frame-cross bracing, etc., were increased; that valve adjustments made are maintained and engines kept square much longer on account of the motion being more direct, rigid and positive for the passage of the valve-driving stresses; and that convenience of the enginemen, inspectors and shop men is promoted by the accessibility of the motion.

There are also available for the use of railroads some patented forms of valve motion or systems of steam distribution which are claimed to obtain economies superior to those of the types already mentioned. The Young valve gear has been applied to engines with Stephenson link and later a Walschaert motion having some detail modifications from the regular design has been proposed. The motion acts upon a wrist plate which has connections to two semi-rotary valves, and the effect of the combination produces valve events remarkable on account of absence of preadmission, small amount of lead, quick port opening, large exhaust area and fine equalization of the events. It is claimed, due to these features, that fine performance and superior economies are obtained which more than compensate for the cost of maintenance of additional parts required.

For the past six years, the Allfree-Hubbell designs for improving steam distribution have been under test and the designers have made changes from time to time as to them seemed best. The first design or "geared system," as the designers called it, has been superseded by what is known as the "compression system," which has been in service for several months. This design is supposed to embrace the economic features of the original design with some additions, and, as it now stands, it attempts to produce the following results: late release, late compression, low clearance, balanced compression, reduced cylinder radiation, quick admission and quick release. In the later design an auxiliary valve has been introduced for the control of compression alone, and allows the main valves to be made and set for a desired release with the Stephenson link, Walschaert, or any other motion, all the changes being made in the cylinders and valve alone.

This device has been under test on several roads and from an average of several reports it seems to give about 6 per cent. reduction in fuel consumption and 5 or 6 per cent. increase in train load, and is able to maintain the same or a little more speed than engines of the same size and ordinary design, except the cylinders. There is also a lighter drain on the boiler for steam, but we are unable to express the amount in figures.

In making a study of this system for causes of the results claimed, we find a very late compression, which, according to the argument of the designers, means that the negative working pressure is acting at a time when it produces the least effect on crank pin. In other words, when the crank pin is very near the center, from this point alone, an increased load is possible, not so much from an increased working pressure as from a reduced negative work. In view of the above argument, any given load could be handled with a less amount of steam, which means a less amount of coal and therefore brings the handling of greater load within the possibilities of the boiler. In addition to this, the reduced clearance is claimed to be responsible for a considerable steam economy.

The report is signed by C. A. Seley (Chairman), R. Quayle, L. H. Turner, J. H. Manning.

Exhibits at the Atlantic City Conventions.

The exhibits at this year's conventions of the Master Car Builders and Master Mechanics Associations have again been placed on the Steel Pier. While the number of exhibitors this year is 285 compared with 254 last year the total floor space occupied is considerably less, 55,500 sq. ft. as against 66,350 sq. ft. last year. The erection and decoration of booths and exhibit spaces has been placed in the hands of one contractor and an artistic and uniform scheme has been worked out. In addition to the exhibitors on the Steel Pier the Goodwin Car Co., Hicks Locomotive & Car Works, Modoc Soap Co., and St. Louis Car Co. have track exhibits near the Pennsylvania Railroad station at the foot of Virginia and Mediterranean avenues. A partial list of the exhibits follows and the remainder will be printed in another list next week.

Adams & Westlake Co., Chicago.—Adlake acetylene gas car lighting system; railroad lanterns; non-sweating, down-draft signal lamps; car hardware, and gas and electric chandeliers. This company also has an exhibit truck, consisting of a demonstration of the Newbold system of electric car lighting from the axle.

Addressograph Co., Chicago.—The card index "Addressograph," combined addressing machine and complete card index system.

American Balance Valve Co., Jersey Shore, Pa.—Balanced slide and piston valves for pressures up to 250 lbs., including piston valves after two years' night and day service passing over ports without bridges; models of the Walschaert valve gear and of a modified Stevens gear, the latter including an internal admission, double-acting Wilson valve at each end of the cylinder.

American Blower Co., Detroit, Mich.—Blowers, ventilators, motor-driven air compressor.

American Brake Shoe & Foundry Co., Mahwah, N. J.—Steel back, wrought lug brake shoes for locomotives, coaches and freight cars.

American File Sharpener Co., New York.—American file sharpening machine; "Carbotivnt" for sharpening files.

American Mason Safety Tread Co., Boston, Mass.—Lead and carborundum safety treads for car steps, stations, etc.

American Steam Gauge & Valve Mfg. Co., Boston, Mass.—Locomotive steam specialties, such as muffled pop safety valves, steam gages, duplex gages, chime whistles, American dead weight gage tester, American Thompson improved indicator, etc.

American Steel Foundries, Chicago.—Cast-steel locomotive frames; cast-steel driving wheel centers; Simplex brake-beam; Simplex driving wheel and coach springs; Simplex car and tender bolsters; cast-steel bolsters; cast-steel side rods and crossheads; Janney cast-steel coupler; the Davis cast-steel car wheel; Susemihl roller-side bearings; Andrews cast-steel side frame; and spring controllers; Kelmont passenger and freight coupler; Leeds pilot coupler and American Steel Foundries Co.'s complete truck mounted in turntable in motion; spring controllers and spring seats.

American Water Softener Co., Philadelphia.—Complete working model of an American water softener.

Anglo-American Varnish Co., Newark, N. J.—Varnishes.

Armstrong Brothers Tool Co., Chicago.—Various designs of tool holders, "Universal" ratchet drills, planer jacks, tool posts, lathe dogs and boring tools.

Atha Steel Castings Co., Newark, N. J.—Cast-steel truck bolsters; "Titan" manganese steel motor gears.

Baeder, Adamson & Co., Philadelphia, Pa.—Large model of refrigerator car, showing hair-felt insulation.

Barnett Equipment Co., Newark, N. J.—Demonstration of the Barnett Connector automatically coupling the steam heat, signal and air-brake service and safety hooks (in lieu of safety chains) between cars; also Barnett universal steam hose coupler.

Besly, Chas. H., & Co., Chicago.—Spiral-grooved steel grinding disks; heavy spiral cloth and paper circles; Helmet tempered taps; Helmet babbitt; Helmet oil, and Helmet bronze, spring and sheets.

Best, W. N., American Calorific Co., New York.—Oil furnaces for annealing, brazing, heating, etc.; also oil burners and safety regulator cocks.

Bethlehem Steel Co., South Bethlehem, Pa.—Staybolt iron, high speed steel, special alloy steel, rails, heavy machinery and special structural shapes.

Bettendorf Axle Co., Davenport, Ia.—Bettendorf 40-ton capacity structural steel underframe equipped with Bettendorf all cast-steel trucks and cast-steel center sill ends; I-beam body and truck bolsters of 30, 40 and 50-ton capacity; trucks of 30, 40 and 50-ton capacity; removable journal box truck frame and riveted arch bar truck frames.

Bickford Drill & Tool Co., Cincinnati, Ohio.—Standard plain radial drill in operation with variable speed motor drive.

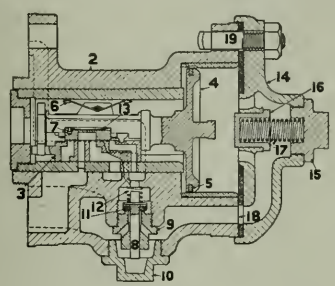
Birdsboro Foundry & Machine Co., Birdsboro, Pa.—Jackson belt lacing ma-

QUICK RECHARGING.

In the graduated release operation the auxiliary reservoir is being recharged through three different ports, instead of one (the feed groove *i*), as formerly, and in such a manner that the rise in auxiliary reservoir pressure is proportional to the fall in cylinder pressure, so that when the release of the brake is complete, the auxiliary reservoir is practically fully recharged and ready for an immediate reapplication if necessary. Having these features incorporated in the automatic system the flexibility of straight-air operation is closely approximated without in any way impairing the safety features of the automatic control. In fact, a distinct advantage from the standpoint of safety is gained by the quick recharge feature, which insures, at all times, the possibility of full braking power being developed, without regard to the length of time intervening after a release—a condition not even approximated by either the straight air or former automatic systems.

TRIPLE VALVES FOR SINGLE CARS.

The quick action features of a triple valve consist of the check valve 15, rubber-seated valve 10 and piston 8, shown



Section Through Type T Triple Valve.

in Figs. 1 to 7. In an emergency application, auxiliary-reservoir pressure is admitted above piston 8, which causes it to unseat valve 10, and allow air from the brake pipe to raise check valve 15 and flow rapidly to the brake cylinder through chambers *Y* and *X* until the brake cylinder and brake-pipe pressures become equalized. The check valve then seats and prevents cylinder pressure from flowing back to the brake pipe. This is the means employed for hastening the brake-pipe reduction throughout the train, as each triple valve vents a portion of the brake-pipe air to its cylinder, which similarly affects the triple valve on the next vehicle, and so on through the train. It also raises somewhat the cylinder pressure obtained, due to the air introduced from the brake pipe. If the brake-pipe reduction is considerably more rapid than the reduction in auxiliary reservoir pressure due to its flow of air to the brake cylinder, the triple-valve piston will assume



Type T Triple Valve.

the emergency position, which will result almost instantly in an application with maximum braking power.

Although the features of graduated release and quick recharging were originally developed in connection with the quick-action triple valve, they can just as readily be incorporated in the plain type of triple valve; that is, one without the quick-action features. This was the next step taken.

PLAIN TRIPLE VALVE WITH GRADUATED RELEASE AND QUICK RECHARGE.

The first form of plain triple valve to be designed to embody the features of graduated release and quick recharge of auxiliary reservoir was for train operation limited to two cars. With this valve the graduated release feature was obtained by piping the exhaust of the triple valve to the brake valve, where the air contained in the brake cylinder would either be retained or allowed to escape to the atmosphere at the will of the motorman. The quick recharging feature was obtained in this valve by means of a small check valve which allowed the brake pipe air to flow directly to the auxiliary reservoir when the triple piston was in release position. A further detailed description of this valve will be unnecessary, as all of its features are embodied in the plain triple valve which followed it, which was designed for use on cars to be operated singly or in trains up to five units. This valve, designated as the type T triple valve, is shown in Figs. 4, 5 and 6.

In Fig. 4, port *a* connects with the brake pipe, port *x* with the reserve supply, port *c* to the brake cylinder, and *p* to the exhaust through the brake valve, while *R* is in connection with the auxiliary reservoir.

Quick recharging is obtained partly through the small check valve shown, which connects *a* through ports *y*, *j* and *u* with the auxiliary reservoir, and partly through ports *x* and *k* from the reserve supply. The graduated release of the brakes is obtained in exactly the same way as previously described for the quick-action triple valve, as will be evident at once from Figs. 5 and 6, the ports being lettered identically. One feature in connection with this valve, however, deserves particular notice. The exhaust port *p* being connected by piping to the brake valve, combined with the use of the reserve supply as previously described, affords two means by which a graduated release of the brakes in the train may be made. The method of graduating the release by means of the reserve supply, in the manner already explained, gives a graduated release on all the cars in the train, while the second method enables the operator to govern the release of the brakes on the head car alone, directly at the brake valve. This latter method makes possible a very flexible control when operating as a single car, and is known as the straight-air release feature.

The application of these valves to trains up to five cars in a service in which frequent stops from a fast schedule were required, made clear the desirability of a more rapid service application of the brakes throughout the train. The modifications already made in the triple valves to obtain the quick recharge feature were easily adapted to produce a quicker service application of the brakes than had previously been possible where the brake-pipe reduction was dependent upon the air contained in the pipe flowing the whole length of the train to the brake valve before it could escape to the atmosphere.

In emergency a rapid serial application of the brakes is obtained by venting a portion of the brake-pipe air to the brake cylinder on each car. This affords a means of escape for the brake pipe air, without all of it having to flow the length of the train to the brake valve. This same method suggested itself for quickening the service application, the action being of the same character as that in emergency, but to a less degree. As there was already a port connecting the brake pipe to the slide valve seat, which in release position affords a means of securing a quick recharge, it was only necessary to add ports to the slide valve, which, in the movement to service position, connect the brake pipe through these ports to the brake cylinder. In this manner a slight reduction in brake-pipe pressure is effected during a service application by each triple valve in turn. By this means the time required to apply the brakes in service is materially reduced.

In Fig. 4, the additional ports just mentioned are marked *o*, *v* and *q*. In the quick service position, port *o* registers with port *y* (see Fig. 7), and cavity *v* in the graduating valve connects ports *o* and *q* in the main slide valve, the latter registering with port *r*, leading to the brake cylinder, thus establishing direct communication from the brake pipe through the check valve to the brake cylinder. When the piston returns to lap position, Fig. 5, the quick service ports *o* and *q* are blanked by the graduating valve. This feature proved to be so easily obtained and of such decided advantage that it was incorporated in all the triple valves designed for use in trains of more than two cars.

Referring to Fig. 8, showing in quick service position the quick-action triple valve already described, it will be observed that the port *q* connects with port *t*, which leads to the top of the emergency piston instead of directly to the brake cylinder, as in the preceding valve. This arrangement was simply a matter of convenience, as the port *t* was already conveniently located for the purpose, and the emergency piston is of sufficiently loose fit to allow the relatively small amount of air thus discharged to pass readily around it and to the brake cylinder without any danger of causing the quick action parts to move.

Referring to Fig. 8, it will be noted that when quick-service port *y* registers with port o in the slide valve, the service ports *d* and *r* only partially register; the air flowing through these ports from the auxiliary reservoir to the bake cylinder is thus restricted in amount, which limits the rate at which the auxiliary-reservoir pressure reduces. If the brake-pipe pressure reduces more rapidly than does the auxiliary-reservoir pressure, the predominance of pressure in chamber *R* will enable the piston to somewhat compress the graduating spring and move the slide valve to the "full service" position shown in Fig. 9. In this position port *y* is closed, prevent-